

(KNN)

Training dataset:

Action Scenes.	Romance Scenes	Happy Ending	Genre
15	2	0	A
1	12	1	R
18	1	0	A
2	14	1	R
14	3	1	A

Testing Data set

Test ID	Action scenes	Romance scenes	Happy ending	Actual Genre
T ₁	10	5	1	A

Comparing:

$$(10, 5, 1) - (15, 2, 0)$$

$$(1, 12, 1)$$

$$(18, 1, 0)$$

$$(2, 14, 1)$$

$$(14, 3, 1)$$

$$(10, 5, 1) (15, 2, 0) = \sqrt{(10-15)^2 + (5-2)^2 + (1-0)^2} = 5.917 \rightarrow A$$

$$(10, 5, 1) (1, 12, 1) = \sqrt{(10-1)^2 + (5-12)^2 + (1-1)^2} = 11.402 \rightarrow R$$

$$(10, 5, 1) (18, 1, 0) = \sqrt{(10-18)^2 + (5-1)^2 + (1-0)^2} = 9.0 \rightarrow A$$

$$(10, 5, 1) (2, 14, 1) = \sqrt{(10-2)^2 + (5-14)^2 + (1-1)^2} = 12.0422$$

$$(10, 5, 1) (14, 3, 1) = \sqrt{(10-14)^2 + (5-3)^2 + (1-1)^2} = 4.4721$$

$$K=3$$

$$\text{Shortest distance} = 5.917; 9.0; 4.473 \rightarrow A$$

$$\text{Scenes weight} = \frac{1}{d} = \frac{1}{5.917} = 0.168$$

$$= \frac{1}{9.0} = 0.11$$

$$= \frac{1}{4.473} = 0.2209$$

$$W = \frac{1}{d} \Rightarrow W_1 = \frac{1}{5.9} = 0.19$$

$$W_2 = \frac{1}{11.40} = 0.08$$

$$W_3 = \frac{1}{9.0} = 0.11$$

$$W_4 = \frac{1}{12.04} = 0.08$$

$$W_5 = \frac{1}{4.47} = 0.22$$

Weight of A.

$$W_1 + W_3 + W_5$$

$$= 0.52$$

$$\text{Weight of A} = W_1 + W_3 + W_5 = 0.52$$

$$\text{Weight of R} = W_2 + W_4 = 0.16$$

$$0.52 > 0.16$$

$$A \gg R$$

Age (x_1)	Previous values (x_2) (in Lakhs)	Car Price (y) in Lakhs
2	3	2
3	4	3
5	1	0.5

Test
 (x_1, x_2, y)
 $(4, 2, 5)$

$(4, 2) \rightarrow (2, 3)$

$\rightarrow (3, 4)$

$\rightarrow (5, 1)$

$$(4, 2) (2, 3) = \sqrt{(4-2)^2 + (2-3)^2} = 2.24$$

$$(4, 2) (3, 4) = \sqrt{(4-3)^2 + (2-4)^2} = 2.24$$

$$(4, 2) (5, 1) = \sqrt{(4-5)^2 + (2-1)^2} = 1.41$$

$K=2 \rightarrow$ Give that in question.

$$d_1 = 2.24$$

$$d_2 = 2.24 \rightarrow 3$$

$$d_3 = 1.41 \rightarrow 0.5$$

$$\hat{y} = \frac{1}{K} \sum_{i=1}^K y_i$$

$$= \frac{3 + 0.5}{2} = 1.75$$

$$= 1.75$$

$$\text{Test} = (4, 4, \textcircled{5}) \rightarrow 1.75$$